Initial Model Training Results

WasteSort | Goal Diggers | BCS34

Actual benchmark executed October 20, 2026. These are measured results from the saved training records and held-out predictions. The later Colab run is a separate execution; its differing CNN scores are not substituted here.

1. Training and held-out comparison

Training: 1,774 images (70.37%); validation: 371 (14.72%); test: 376 (14.91%). Seed 42; candidate duplicate groups remain within one partition. Selection used validation macro F1, with deployment frozen before test evaluation.

Model	Validation macro F1	Test accuracy	Test macro F1
HOG + SVM	60.62%	65.96%	63.52%
HOG + Random Forest	55.94%	63.56%	60.22%
Small CNN	62.07%	59.57%	57.68%
MobileNetV2	84.74%	84.57%	81.22%
Weighted CNN (ablation)	60.32%	58.78%	58.84%

MobileNetV2: 318/376 correct predictions; macro precision 82.05%, macro recall 80.58%, macro F1 81.22%. Macro averages weight all six classes equally.

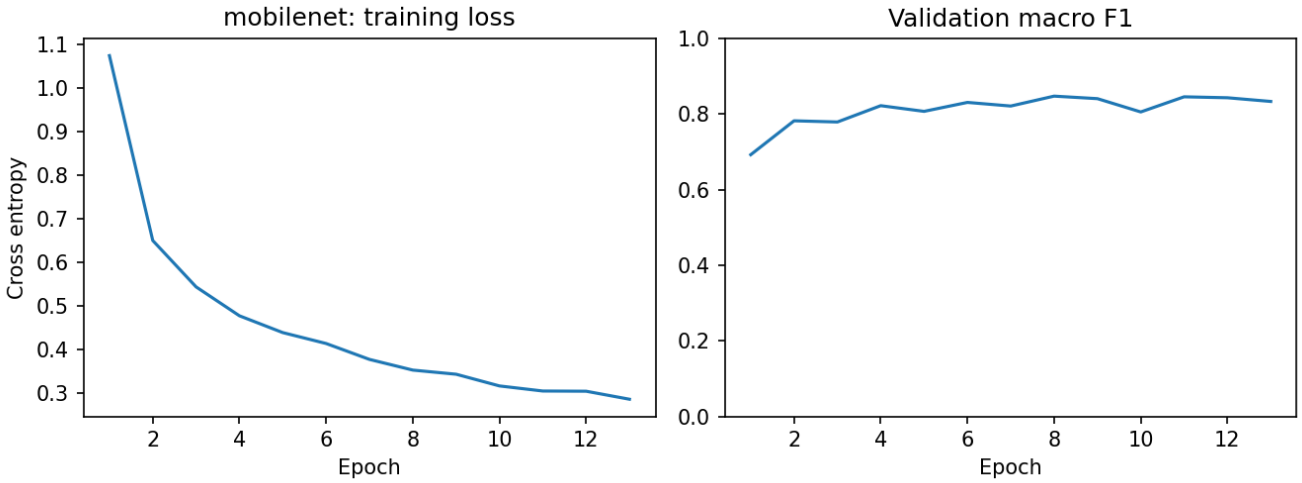
2. Selected-model class performance

Class	Precision	Recall	F1	Test n
cardboard	96.30%	88.14%	92.04%	59
glass	80.82%	80.82%	80.82%	73
metal	81.25%	82.54%	81.89%	63
paper	86.14%	93.55%	89.69%	93
plastic	85.29%	82.86%	84.06%	70
trash	62.50%	55.56%	58.82%	18

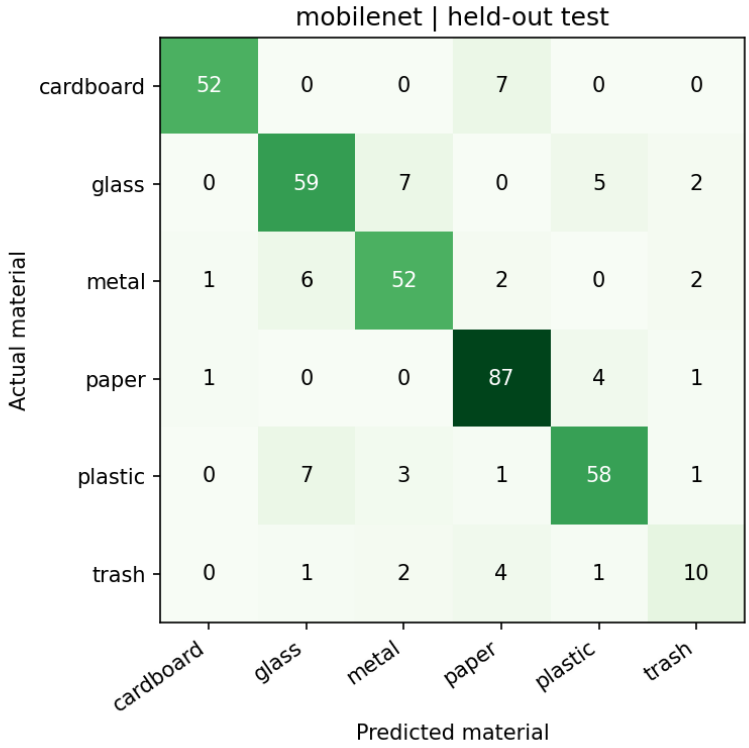
The model found only 10 of 18 trash images (55.56% recall), missing eight. Stronger results on larger classes can hide this weakness. One fitted run and split support preliminary feasibility, not campus accuracy or formal statistical superiority.

Evidence: test_evaluation.json, test_predictions.csv, comparison.csv and mobilenet_training.json in 03_Results_Evidence. Introductory KNN/Decision Tree validation results and the separate numerical regression exercise are also supplied there.

Training behavior and classification errors



The saved learning curve shows training loss and validation macro F1 over epochs. Best checkpoint: epoch 8; the pretrained backbone was frozen and a six-class head was trained. Early stopping used validation performance, never the test score.



Confusion matrix: true material by row and predicted material by column. The diagonal counts correct predictions. Off-diagonal counts show where errors occur; they should guide a separately planned follow-up, not repeated tuning to this exposed test set.

Separate cloud check: the full Colab notebook completed all six code cells on October 26. MobileNetV2 displayed the same rounded test metrics, while CNN results differed. The output notebook is included; the cloud experiment ZIP was not recovered, so exact checkpoint equivalence is not claimed.